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AuxFlow: Anchor-grounded homography estimation through flow-guided auxiliary points for Soccer field registration and player localization ☆

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ABSTRACT

We introduce AuxFlow, a novel, temporally-aware pipeline for homography estimation and field registration in challenging football broadcast footage. To overcome the temporal instability and high performance variance of existing per-frame keypoint methods, our AuxFlow approach combines a robust frame-wise keypoint model with a temporal propagation strategy. The system automatically identifies high-confidence "anchor" frames where it estimates the homography solely based on the keypoint model, before sampling auxiliary points, which are re-identified in neighbouring frames using optical flow to establish dense, coherent correspondences across the sequence. This significantly enhances the stability and accuracy of the estimated homographies. Our evaluation on the SoccerNet GSR dataset shows consistent, measurable improvements in robustness and smoothness over existing State-of-the-Art, enabling highly reliable player localization invaluable for downstream applications.

1. Introduction

Computer vision has been successfully applied to a wide range of real-world problems, enhancing the ability to analyse and interpret complex scenes. In sports, and in particular in team sports as, e.g., soccer, basketball and ice hockey, recent advances in deep learning based image and video understanding have provided new insights for players, coaches, analysts, and spectators (Li et al., 2025; Fujii, 2025; Thomas et al., 2017; Mgaya et al., 2021). The popularity of these methods can be attributed to the generally straightforward access to video data, either through sports broadcasts, which are often readily available at the scene, or by conducting video recordings with relatively uncomplicated setups.

Due to the high commercial and public interest in sports broadcasting, professional matches have long been recorded with single-camera as well as multi-camera setups. In the single-camera setting, the footage typically consists of panned and zoomed views designed to follow the action across the entire field. Although these recordings are usually suited for intuitive understanding on a qualitative level, quantitative analyses such as measuring total distances of players, speeds, positioning, and numerous others, are not feasible based on the raw video perspective. This, however, is particularly interesting when it comes to automated or at least assisted analysis techniques.

Many downstream tasks, including tactical analysis, player tracking, and event recognition, require accurate localization of athletes in real-world field coordinates rather than only in image coordinates. Achieving this requires a mapping between the broadcast image and the playing surface, a task commonly referred to as field registration. In practice, this involves calibration of the camera setup.

In multi-camera systems, calibration is facilitated by the availability of multiple perspectives. In contrast, single uncalibrated broadcast cameras require the exploitation of the known geometry of the playing surface, i.e. field line markings according to standardized rules of the particular sport. Since both the image plane, according to the pinhole camera model, and the field plane are two-dimensional manifolds in a common three-dimensional space, their relationship can be modelled as a homography. This mapping is fully defined by four point correspondences, although the use of additional points typically leads to more stable and robust solutions.

Early approaches for field registration were based on classical computer vision techniques that detected field lines and markings using, e.g., edge detectors or hand-crafted features such as SIFT or ORB (Rublee et al., 2011; Lindeberg, 2012). With the development of deep learning, two major strands of research emerged. The first focuses on direct regression of camera parameters or homography matrices, for instance with encoder–decoder architectures or retrieval-based

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methods from annotated scene databases (Shi et al., 2022). The second direction uses deep models to detect semantic field points, i.e. keypoints such as intersections, lines, or circle tangents with higher reliability than classical detectors (Theiner and Ewerth, 2023; Homayounfar et al., 2017a). The identification of global features, including the halfway line or tangent points of the centre circle, has proven particularly effective since these markings provide stronger geometric constraints (Nie et al., 2021; Gutiérrez-Pérez and Agudo, 2024).

Current state-of-the-art methods for semantic feature detection achieve very high accuracy on frames where sufficient field markings are visible. However, their performance degrades substantially when fewer markings are present, as has been documented for example on the SoccerNet-GSR benchmark (Somers et al., 2024; Magera et al., 2025). Recent research has therefore turned to temporal information in video sequences, rather than treating each frame independently.

Temporal consistency is an essential feature for movements on the playing field and can therefore be exploited to stabilize calibration, thereby improving accuracy both in well-observed and in challenging frames. Importantly, this consistency may not be clearly visible in the camera view, as changes in a player's position are subject to both their actual movement on the field and the movement of the camera. Previous approaches include interpolation of homography matrices, probabilistic filtering techniques, and more recently the propagation of field correspondences by means of optical flow (Magera et al., 2025). While our method also focuses on propagating such correspondences, it uses spatial feature sampling techniques rather than relying on a fixed set of keypoints provided by the keypoint detection model. It also incorporates a semantic segmentation model for regular re-initialization.

As with previous methods, the main objective of this work is to enhance homography estimation in non-ideal frames, i.e. frames in which thorough semantic field correspondences cannot be established. We propose to sample auxiliary field points in good frames (i.e., frames where a homography can be detected reliably from visible semantic keypoints) and track them through optical flow to improve homography estimation in non-ideal frames. We therefore test multiple sampling techniques including regular sampling, random sampling and informed sampling using additional geometric priors. We establish field correspondences by projecting these samples to pitch coordinates and finally propagate the sampled keypoints to adjacent frames. The propagated correspondences serve to enrich the homography estimation process, enforce temporal consistency, and transfer information about otherwise unobservable features. In our experiments, classical Lucas–Kanade flow (Lucas and Kanade, 1981) is compared with the learned RAFT model (Teed and Deng, 2020). Finally, a homography quality assessment method is proposed in which warped field masks are compared with U-Net-based predictions of the playing surface.

The contributions of this work can be summarized as follows. First, a tracking scheme is introduced that propagates field features (i.e., auxiliary points) via optical flow to improve homography estimation under non-ideal conditions. Second, spatial sampling strategies for auxiliary points are analysed with respect to robustness and accuracy. Third, a comparison between classical and learned optical flow approaches in the given setting is provided. Fourth, novel heuristic initialization procedures for selecting near-optimal starting frames are proposed. Finally, a homography quality assessment strategy based on U-Net segmentation is presented that helps to identify good starting (anchor) frames to begin propagation. The scope of this work is limited to soccer broadcast footage, with examples drawn from the GSR dataset. The complete codebase is openly accessible to the research community[1].

[1] Code available at https://github.com/juliantziegler/AuxFlow.

2. Related works

2.1. Field marking detection

The estimation of a homography fundamentally depends on the identification of sufficient correspondences in the planes to be registered. Early work focused on finding well-defined field markings, where the corresponding real coordinates were known due to the set of rules defined for specific sports. These include using Hough line transforms or binary segmentation of the field (Farin et al., 2003; Dubrofsky and Woodham, 2008; Alemán-Flores et al., 2014). More recently, the detection or semantic segmentation of the field markings can be quite well executed using deep neural networks, most often with an encoder–decoder structure (Theiner and Ewerth, 2023; Sha et al., 2020; Nie et al., 2021). In our work, we will utilize the open source SOTA keypoint detection model NBJW (Gutiérrez-Pérez and Agudo, 2024) to detect initial correspondences.

2.2. Augmented correspondences

As homography quality is dependent on the number of well localized correspondences, the accuracy of these methods varies significantly from view to view, as the number of field markings are lower in some regions compared to others (Claasen and Villiers, 2024; Magera et al., 2025). This led to the development of multiple schemes to extract additional correspondences through augmentation. Early examples are the derivation of vanishing points or the extension of lines to generate new intersections (Hayet et al., 2005; Homayounfar et al., 2017a; Cuevas et al., 2020; Falaleev and Chen, 2024; Gutiérrez-Pérez and Agudo, 2024; Tsurusaki et al., 2021). Using circular markings for homography estimation involves the use of circle equations, which render the process more computationally heavy and less accurate (Gutiérrez-Pérez and Agudo, 2024; Claasen and Villiers, 2024). The use of tangents to specifically annotate and train specific points of these circular markings delivered more high quality correspondences while getting rid of the circle equations (Andrews et al., 2024; Cuevas et al., 2020; Falaleev and Chen, 2024; Gutiérrez-Pérez and Agudo, 2024). These works all use real field markings, or extensions thereof, for correspondence generation. Other possibilities include the detection of lines formed by the grass mowing pattern for additional intersection points (Cuevas et al., 2022). However, this method naturally relies on the existence of these mowing patterns, which cannot be guaranteed to exist on any pitch. This is also the case for a large part of the GSR dataset (Somers et al., 2024) used in this study. Some authors have trialled annotating their dataset with a regular grid of points and learning a deep network to predict these based on global localization instead of local features (Chu et al., 2022; Claasen and Villiers, 2024; Maglo et al., 2023, 2022; Nie et al., 2021). However, to our knowledge, augmentation based on the field markings still delivers SOTA in the form of the NBJW model.

2.3. Direct homography estimation

As pointed out earlier, apart from methods which try to derive correspondences to then estimate the homography, other approaches have been explored to derive a homography from the frame. These include direct regression of the homography matrix with a deep neural network (Shi et al., 2022; Jiang et al., 2020; Fani et al., 2021). The results show less overall accuracy than competing marking detectors. The same can be said for look-up-table methods, which store homographies and corresponding camera views, or feature space representation thereof, and based on the camera view seen in the frame interpolate from their database to predict a transformation (Chen and Little, 2019; D'Amicantonio et al., 2023; Sha et al., 2020; Sharma et al., 2017; Tarashima, 2021; Zhang and Izquierdo, 2021). These, too, fall short of SOTA and are also less computationally efficient and lack interpretability.

2.4. Tracking and temporal consistency

Until recently, datasets with broadcast video footage with annotated calibration or homography data were of limited availability. As a result, only a limited amount of research specifically towards sports field registration in video data has been conducted. This data type emphasizes the temporal consistency of homography estimation and localization across all frames.

One possible approach is to calculate the homography between two consecutive frames, or rather the homography that maps one view of the sports field to the next. This is done by identifying correspondences, either points or lines, between frames and estimating the homography using a robust estimator like RANSAC. A range of methods apply common tracking filters, such as Claasen and Villiers (2024), who used a Kalman Filter to track correspondences between frames. Others have used an extended Kalman filter to track the camera parameters (Beetz et al., 2007), while Citraro et al. (2020) make use of a particle filter to model the motion of the camera.

More recent work has focused on online, video-wise refinement. For example, BroadTrack (Magera et al., 2025) proposes a full camera calibration pipeline that builds upon the semantic segmentation of TVCalib (Theiner and Ewerth, 2023). In each subsequent frame, it builds a multi-part loss function and optimizes the camera parameters using Levenberg–Marquardt. This loss function includes a field marking loss from points sampled by TVCalib, an optical flow loss from tracked points, and a tripod constraint loss.

While our method also utilizes optical flow for temporal consistency, it is fundamentally distinct from approaches like BroadTrack in several key aspects. Firstly, we focus on the crucial task of identifying the best frames for initialization and reinitialization, respectively. We do this through heuristics based on the confidence scores from the NBJW keypoint model (Gutiérrez-Pérez and Agudo, 2024) and a novel approach with a U-Net-based surrogate metric. Secondly, our approach to optical flow is based on a comparison of various point sampling strategies (uniform, random, and geometric) on the 2D pitch coordinates. This ensures a more balanced and representative set of tracked points compared to BroadTrack's method of sampling from the image plane, which can lead to uneven point distributions on the field. By using the homography from the previous frame to project known points to the current frame, we directly derive their pitch coordinates and integrate them into our RANSAC-based homography refinement. This unique methodology leverages the strengths of both static feature detection and temporal tracking to achieve robust and accurate field registration.

2.5. Playing field segmentation

Playing field segmentation serves as a fundamental, low-level primitive in computer vision for sports analytics, enabling subsequent high-level tasks like player tracking, metric analysis, and Game State Reconstruction (GSR) (SoccerNet, 2024; Berjón et al., 2021; Deng et al., 2024). Traditional Computer Vision methods relied on colour-space analysis, such as Hue-based or RGB thresholding, assuming a distinct green field colour (Niu et al., 2012; Homayounfar et al., 2017b). However, these approaches lacked the robustness required for automated systems, often failing under real-world conditions due to high sensitivity to photometric variability, such as shadows and uneven lighting, which made automatic threshold selection unreliable (Niu et al., 2012; Xu, 2024). The field shifted with the integration of Deep Learning, which provided the necessary feature invariance (Deng et al., 2024). While earlier DL models like Fully Convolutional Networks were effective for large-scale segmentation of the field surface, they often failed to resolve the fine boundaries of thin structures. The U-Net architecture, distinguished by its symmetric encoder–decoder structure and crucial skip connections, became preferred for its ability to retain high-resolution feature maps. This architectural choice ensures the high-fidelity segmentation of critical geometric primitives, such as the thin pitch lines and circle arcs (Xie and Li, 2018). The primary utility of this precise segmentation is to facilitate homography estimation and camera calibration, which maps the 2D image plane onto a standardized 2D metric model of the playing field (Alemán et al., 2024; Homayounfar et al., 2017b). Benchmarks like SoccerNet reflect this focus, defining success not merely by pixel accuracy but by the precise localization of pitch line extremities and goal posts, as errors in segmentation propagate and ultimately constrain the accuracy of all downstream 2D metric estimates derived during GSR (SoccerNet, 2024; Golovkin, 2024). Despite progress, challenges persist in managing dynamic occlusion from players in high-speed sports like basketball (Roboflow, 2023), ensuring cross-sport generalization (Deng et al., 2024), and optimizing model architectures for real-time computational efficiency (Maurer, 2025).

3. Methods

3.1. Overview

We introduce a novel homography estimation pipeline (Fig. 1 tailored to broadcast video sequence data, such as the SoccerNet GSR dataset (Somers et al., 2024). Our approach builds upon a set of world-to-image correspondences $\mathcal{K}_i = \{(\mathbf{x}_k, \mathbf{x}'_k)\}$, $k \in \mathbb{N}$ obtained from a keypoint detection model (in our case, NBJW (Gutiérrez-Pérez and Agudo, 2024)), where $\mathbf{x}'_k$ denotes the image coordinate of a semantic keypoint corresponding to the world coordinate $\mathbf{x}_k$. From the set of all frames $\mathcal{F}$, we select a set of anchor frames $\mathcal{A} = \{f_j^A\}$, $j \in \mathbb{N}$, $\mathcal{A} \subset \mathcal{F}$, which are assumed to exhibit high-quality homographies. Anchors are determined based on heuristic rules or surrogate metrics (see Section 3.2).

For each anchor frame f_j^A, we estimate the homography H_j^A from the set of correspondences in $\mathcal{K}_j$. Starting from each anchor f_j^A, we begin the iterative propagation procedure to acquire homographies for all frames $\mathcal{F} \setminus \mathcal{A}$, $|\mathcal{A}| < |\mathcal{F}|$. This process involves three core steps performed for each frame-pair $f_i, f_{i\pm1}$. For readability reasons and without loss of generality, we will stick with the notation f_{i+1} to denote any frame $f_{i\pm1}$ adjacent to frame f_i from hereon. Given an estimated homography H_i for frame f_i, the steps are as follows:

- **Auxiliary Point Sampling:** We sample a set $\mathcal{P}_i$ of auxiliary point correspondences by projecting a world point $\mathbf{x}_i$ in the current frame f_i to its corresponding image coordinate $\mathbf{x}'_i$ using H_i, i.e. $\mathbf{x}'_\mathbf{i} = \mathrm{H}_i^{-1}\mathbf{x}_i$. Several strategies for sampling these points are explored (see Section 3.3).
- **Optical Flow:** Given $\mathcal{P}_i$, the corresponding image points $\mathbf{x}'_{i+1}$ in the adjacent frame f_{i+1} are identified using optical flow between f_i and f_{i+1} to build $\mathcal{P}_{i+1}$. As the corresponding world points do not change from one frame to another, their coordinates are given by $\mathbf{x}_{i+1} = \mathbf{x}_i$. Furthermore, we explore both a classical and deep learning driven approach to optical flow estimation (see Section 3.4).
- **Homography Estimation:** The new homography H_{i+1} is then estimated, leveraging both auxiliary correspondences $\mathcal{P}_{i+1}$ and the keypoint correspondences $\mathcal{K}_{i+1}$, ensuring a more robust result based on more information (cmp. Section 3.5).

With our method now outlined, we provide detailed descriptions of each step.

3.2. Anchor frame selection

The automated selection of good anchor frames is key to our method performing well, as all homographies estimated through propagation are, to an extent, dependent on the anchor frame homography. We introduce both a heuristic approach, considering the correspondences $\mathcal{K}_i$ from the keypoint method, and two surrogate metrics that allow us to quantify the homography quality using a U-Net trained for binary pitch segmentation.

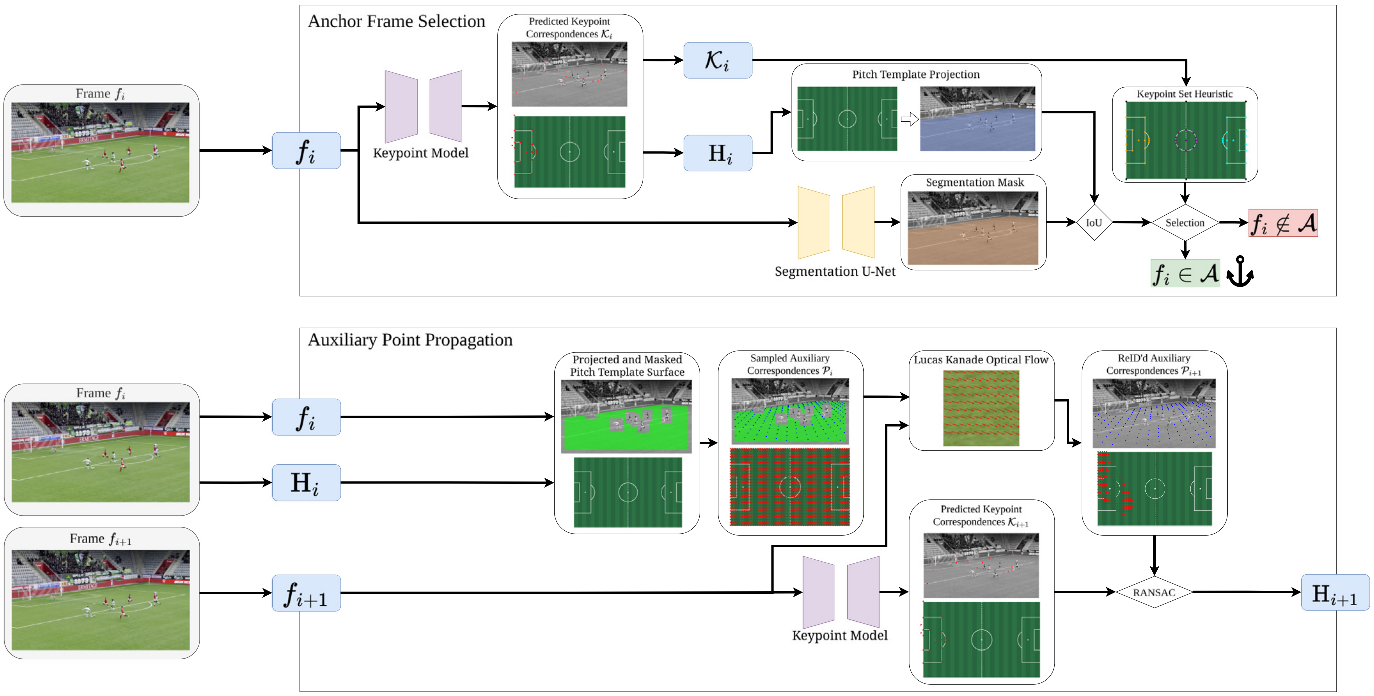


Fig. 1. Methodology overview. As can be seen in the top half, anchor frames are selected globally by examining IoU of the projected pitch template and the estimated pitch segmentation, along with a keypoint heuristic. For these anchor frames, keypoint model estimates for homographies are used. Propagation across neighbouring frames uses both the keypoint correspondences $\mathcal{K}_{i+1}$ from the NBJW method and the auxiliary correspondences $\mathcal{P}_{i+1}$ generated by projecting and re-identifying sampled points to estimate the next homography, as shown in bottom half. Best viewed on screen.

(a) Good Frame (Pitch Area) (b) Bad Frame (c) Good Frame (Centre Circle Area)

Fig. 2. Comparison of detected field marking points (blue) by NBJW method. Figs. 2(a) and 2(c) show examples of regions where we expect to find many keypoints using NBJW, which we assume lead to good homography estimates. Fig. 2(b) on the other hand shows an example frame where almost no keypoints can be detected by NBJW. We expect the estimated homography to be of lower quality. Best viewed on screen.

3.2.1. *Pitch view heuristic*

Our heuristic is based on the assumption that a high number of detected field-marking correspondences correlates with a high-quality homography estimation. One would expect NBJW to produce substantially more accurate homographies in frames such as Figs. 2(a) and 2(c), than in a frame such as Fig. 2(b). Empirically, three pitch regions consistently yield sufficient keypoints for reliable estimation: the centre circle and the two penalty areas, denoted by the keypoint sets $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$, shown in Fig. 3. The analysis in Table 1 supports this hypothesis, showing that frames containing at least one of these regions exhibit significantly higher mean and median $\mathrm{IoU}_{\mathrm{entire}}$ values (see Section 4.2) compared to all other frames, which show markedly lower accuracy when none of the sets are fully present. Based on this observation, we define our heuristic for anchor frame selection as

$$f_i \in \mathcal{A} \quad \text{if} \quad \exists r \in \{1,2,3\} : \mathcal{S}_r \subseteq \mathcal{K}_i, \tag{1}$$

i.e., a frame f_i is selected as an anchor if its keypoint set $\mathcal{K}_i$ fully contains one of the predefined keypoint sets $\mathcal{S}_1$, $\mathcal{S}_2$, or $\mathcal{S}_3$ (see Fig. 4).

3.2.2. *U-Net segmentation surrogate metrics*

To evaluate the accuracy of homographies estimated using keypoints in $\mathcal{K}_i$ without relying on ground truth labels, we design surrogate

Table 1
Mean and median $\mathrm{IoU}_{\mathrm{entire}}$ (cmp. Section 4.2) for frames containing each region, versus all other frames.

Region	Mean $\mathrm{IoU}_{\mathrm{entire}}$ ↑	Median $\mathrm{IoU}_{\mathrm{entire}}$ ↑
All frames	0.7351	0.8270
Right penalty area ($\mathcal{S}_1$)	0.8658	0.9050
Centre circle ($\mathcal{S}_2$)	0.8342	0.8621
Left penalty area ($\mathcal{S}_3$)	0.8350	0.8553
Remaining frames	0.5928	0.7027

metrics that compare a pitch model against the segmentation output of a U-Net. The intuition is that if the estimated homography is close to the ground truth, then the canonical pitch template $\mathcal{T}$ reprojected into the image should align well with the visible pitch segmentation.

Surrogate metric 1 — image-space mask IoU:. The first surrogate metric is defined as the intersection over union (IoU) between two binary masks $\mathcal{M}_{\mathrm{seg}}$ and $\mathcal{M}_{\mathrm{reproj}}$, where

- $\mathcal{M}_{\mathrm{seg}}$ is the pitch mask predicted directly by the U-Net, and

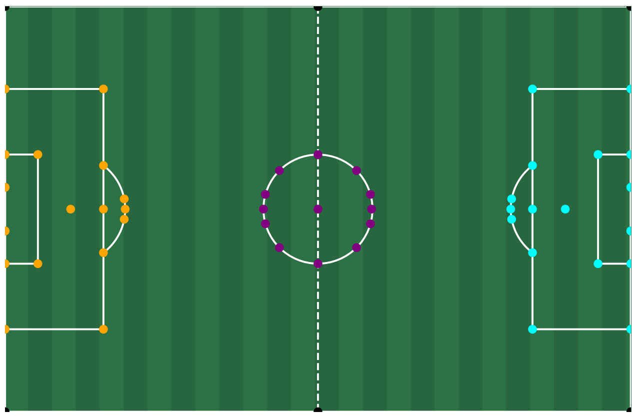
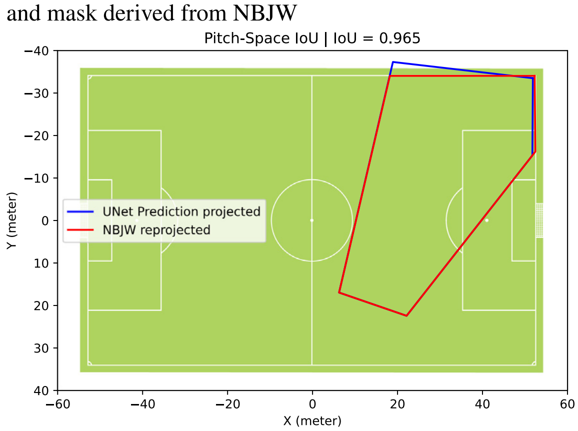


Fig. 3. Figure shows each of the three keypoint sets used for the anchor frame selection heuristic. Right penalty area, $\mathcal{S}_1$, is shown in cyan, centre circle ($\mathcal{S}_2$) in purple and the left penalty area $\mathcal{S}_3$ in purple. Best viewed on screen.

(a) Surrogate Metric 1: IoU between predicted pitch mask and mask derived from NBJW



(b) Surrogate Metric 2: IoU between reprojected NBJW in pitch plane and reprojected predicted mask

Fig. 4. Visualization of surrogate metrics. In both cases, we use IoU to compare the predicted mask of the U-Net with the mask projected by NBJW. Metric 2 however reprojects both masks into the pitch plane using H_i before calculating IoU. Best viewed on screen.

- $\mathcal{M}_{\text{reproj}}$ is the mask obtained by reprojecting the canonical pitch template $\mathcal{T}$ into the image using the homography H_i^{-1}.

In case the current homography H_i approximates the ground truth homography H_{gt} sufficiently well, i.e. $H_i \approx H_{gt}$, we can expect that the predicted mask $\mathcal{M}_{\text{seg}}$ coincides with the reprojected mask $\mathcal{M}_{\text{reproj}}$, eventually resulting in an IoU score close to 1. In practice, we propose selecting anchor frames by identifying local maxima in this score along the video sequence. A key limitation of this metric is its reliance on image-space comparisons: due to perspective distortion in broadcast camera views, a fixed physical error on the pitch corresponds to a much larger pixel error near the bottom of the image than near the top. This spatial bias reduces the reliability of the measure.

Surrogate metric 2 — pitch-space mask IoU:. To mitigate the spatial bias, we introduce a second metric computed in pitch coordinates. Both masks are first transformed into the canonical pitch space using H_i, after which the IoU is evaluated. This compensates for perspective distortions and provides a more uniform assessment of homography quality across the field.

3.3. Auxiliary point sampling

In contrast to other methods described in literature, our method aims to add additional correspondences apart from the field markings to the set of points used to estimate the homography. We consider multiple approaches to sample the set of auxiliary correspondences $\mathcal{P}_i$ in a given frame f_i.

For all methods, we first ensure sampling only occurs in the area of the canonical pitch template that is actually visible. Therefore, given the current frame f_i and its corresponding homography H_i, the mask

$$\mathcal{M}_{\text{pitch}}(\mathbf{x}_i') = \begin{cases} 1 & \text{if } H_i\mathbf{x}_i' \in \mathcal{T} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

contains all image pixels constituting the pitch in the current view. We enforce that no pixels belonging to moving objects are sampled by excluding all pixels covered from any of the players bounding boxes or the balls bounding box by defining the mask

$$\mathcal{M}_{\text{occ}}(\mathbf{x}_i') = \begin{cases} 0 & \text{if } \mathbf{x}_i' \in \bigcup_{z=1}^{Z} \text{pad}(B_z, 30) \\ 1 & \text{otherwise} \end{cases} \tag{3}$$

where B_z denotes the zth of Z bounding boxes, and pad$(\cdot)$ expands the bounding box in each dimension by 30 pixels. Moreover, we ensure that the sampled points are re-identifiable in the next frame by excluding points at the image boundaries. To this end, we define the mask

$$\mathcal{M}_{\text{bor}}(\mathbf{x}_i') = \begin{cases} 0 & \text{if } \mathbf{x}_i' \notin [b, W-b] \times [b, H-b] \\ 1 & \text{otherwise} \end{cases} \tag{4}$$

with W the image width, H the image height and b the amount of border pixels not considered for sampling, which we set to 50.

Both padding parameters are necessary to forego sampled points either being occluded by moving players, being on players themselves, or not contained in the consecutive frame. This would drastically degrade propagation. We found the maximal flow of a pitch point between two images in the training set to be approximately 30 pixels. Therefore, we set the padding of the player bounding boxes to that value. For the image edge, we applied additional tolerance and used 50 pixels, to reduce anomalies. For other datasets or resolutions, a similar investigation should reveal the necessary padding.

The final binary mask, which we apply to all of our sampling strategies, defines the valid image area from which auxiliary points can be sampled and is given by:

$$\mathcal{M}_{\text{val}}(\mathbf{x}_i') = \mathcal{M}_{\text{pitch}}(\mathbf{x}_i') \cap \mathcal{M}_{\text{occ}}(\mathbf{x}_i') \cap \mathcal{M}_{\text{bor}}(\mathbf{x}_i').$$

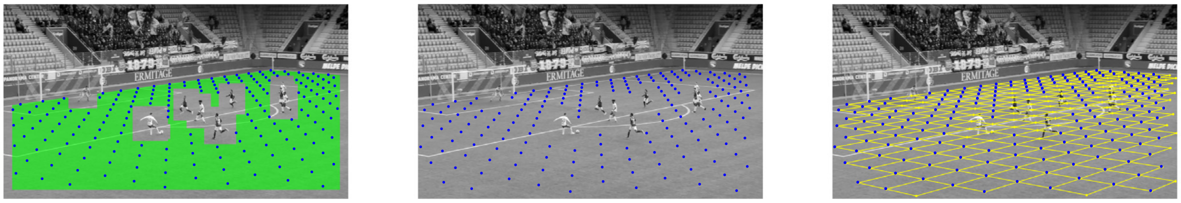
In the following, we present the sampling strategies we implemented.

3.3.1. Random sampling

This baseline method samples a predefined number of image points $\mathbf{x}_i' \in \mathbb{R}^2$ uniformly at random from the set of valid mask locations.

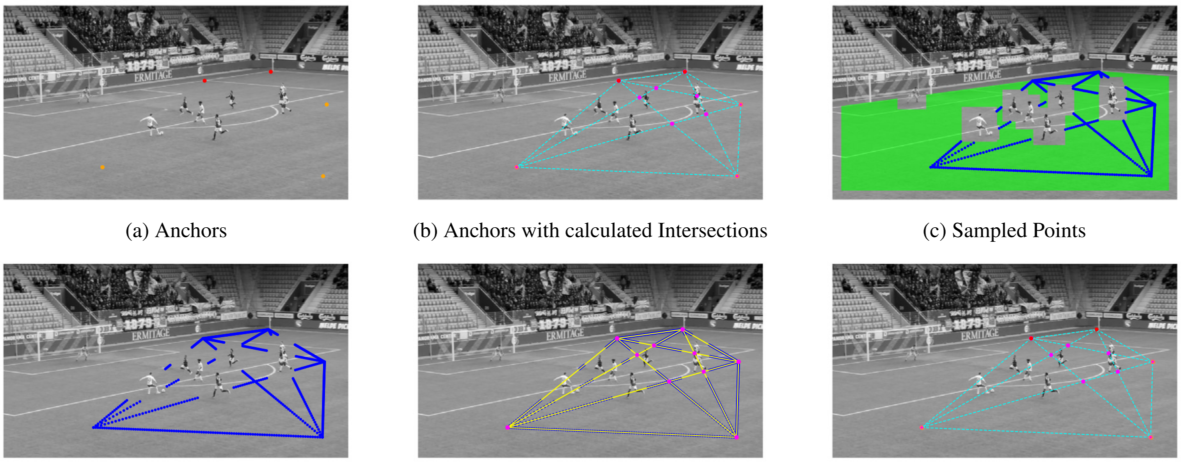
$$\mathbf{x}_i' \sim \text{Uniform}\left(\{\mathbf{x} \mid \mathcal{M}_{\text{val}}(\mathbf{x}) = 1\}\right) \tag{5}$$

The corresponding world points $\mathbf{x}_i$ are calculated using H_i. This is similar to the method presented by Magera et al. (2025) as part of their camera parameter optimization scheme.



(a) Auxiliary Point Sampling (b) Re-Identification (c) Post-Processing with RANSAC along Grid Lines

Fig. 5. Visual representation of a propagation step. (a) In frame f_i, using H_i, we mask the image according to the visible pitch area (green). The regular grid of world points is then projected into the image using H_i^{-1}, and filtered by the mask. The resulting image points of the auxiliary correspondences $\mathcal{K}_i$ are depicted in blue. (b) Going on to frame f_{i+1}, the image points are re-identified (blue). This builds the auxiliary point set $\mathcal{K}_{i+1}$. (c) If enabled, we use a line RANSAC post-processing step, shown in yellow, to refine the estimates. Best viewed on screen.



(a) Anchors (b) Anchors with calculated Intersections (c) Sampled Points

(d) Re-Identified Points (e) RANSAC Reconstruction (f) Reconstructed Correspondences

Fig. 6. Visual overview of the Geometric Point Sampling process. First, NBJW (red) and random (orange) anchors are selected (a). Intersections (magenta) and connecting lines (cyan) are calculated between anchor pairs (b). Next, points (blue) are sampled along these lines, masked by the pitch template (green) to exclude padded player detections and borders (c). These points are re-identified in the subsequent frame (d). Finally, the geometry is reconstructed by assembling lines via RANSAC (e) and determining new image coordinates for valid intersections (f). Best viewed on screen.

3.3.2. Regular grid sampling

In this strategy, a regular grid of points is constructed on the pitch template in world coordinates at intervals of 2.5 and 2 m along the x- and y-axes, respectively, see Fig. 5(a). Let $\mathbf{x}_i \in \mathbb{R}^2$ be a world point on this regularly sampled grid. This point is then projected to the image plane using the homography H_i^{-1} and only retained if its projection falls within the valid mask $\mathcal{M}_{\mathrm{val}}$, i.e.

$$\mathbf{x}_i' = \mathrm{H}_i^{-1}\mathbf{x}_i, \quad \forall\ \mathbf{x}_i \text{ where } \mathcal{M}_{\mathrm{val}}(\mathrm{H}_i^{-1}\mathbf{x}_i) = 1. \tag{6}$$

In cases where the projection yields noninteger pixel values, the nearest neighbouring pixel is used.

3.3.3. Geometric sampling

This strategy leverages geometric priors to sample points along lines which define intersections. To these intersections, we calculate the correspondences. In theory, re-identifying the points and reconstructing the lines should be more robust to noise than simply re-identifying points. A visual description is given in Fig. 6.

The process begins with two high-confidence keypoints sampled in the image space from $\mathcal{K}_i$, along with three randomly selected pitch points sampled for maximum pitch coverage. This is achieved through sampling of 25 points and choosing those which cover the largest area. We refer to these points as primary anchor points. A set of lines is formed by connecting all possible pairs of these primary anchor points. The intersections of these lines constructed from primary anchor points define secondary anchor points in the image plane. These intersections are the points from which correspondences will be determined later on. This method ensures that sampled auxiliary points are established via the intersection of at least two lines. In contrast, random sampling and regular grid sampling as introduced above use singular sampled points.

Given that lines are formed as described above, 100 image points are uniformly sampled from each line, see Fig. 6(c). This sampling strategy is later used in conjunction with line RANSAC to re-identify points in the next frame, as described in Section 3.4.2. Corresponding world points are retrieved using H_i.

3.4. Re-identification of correspondences

3.4.1. Optical flow tracking

In this step, as previously outlined, the goal is to find the image points in frame f_{i+1} which share real world correspondences with the image points in frame f_i. More precisely, given an image point $\mathbf{x}_i'$ in the current frame f_i with real world correspondence $\mathbf{x}_i$, the goal is to find the image point $\mathbf{x}_{i+1}'$ in the adjacent frame f_{i+1} that shares the real world correspondence $\mathbf{x}_i$. We assume this to be possible by estimating the flow between the two frames f_i and f_{i+1}. An example of re-identified correspondences is given in Fig. 5(b).

Given a point $\mathbf{x}_i' \in \mathbb{R}^2$ in frame f_i, the corresponding location in frame f_{i+1} is estimated via the local flow vector $\mathbf{v}_i = \mathrm{flow}(\mathbf{x}_i')$:

$$\mathbf{x}_{i+1}' = \mathbf{x}_i' + \mathbf{v}_i. \tag{7}$$

Furthermore, for more robust results, we employ forwards–backwards tracking, in which we also estimate the reverse flow $\mathbf{v}_i'$ from f_{i+1} to f_i.

Table 2
Location Similarity (LocSim) and median distance of player positions when using a homography derived from ground-truth field markings.

Metric	LocSim		Distance (m)
	Mean	Median	Median
H^{gt}	0.9928	0.9992	0.08

This allows us to derive a tracking error φ given by

$$\varphi = \|\mathbf{v}_i + \mathbf{v}'_i\|_2 \ ,$$

which corresponds to the Euclidean distance between the image point $\mathbf{x}'_i$ and the resulting point after applying both forward and backward flow. Using φ as a thresholding parameter, only correspondences with $\varphi \leq 1$ when using Lucas–Kanade or $\varphi \leq 3$ when using RAFT are placed into $\mathcal{P}_{i+1}$. Thresholds were derived empirically (cmp. Appendix).

We examine two methods to obtain the flow within this work. As a baseline method, we use Lucas–Kanade optical flow in its pyramidal implementation (Lucas and Kanade, 1981). Also, we train a custom implementation of the RAFT (Teed and Deng, 2020) architecture on patches of the SoccerNet GSR dataset, which we annotated using the ground truth homographies. The trained model was used in inference on small 80×80 pixel sized patches around the sampled points, where it predicts the local flow $\mathbf{v}_i$. At the heart of RAFT is a Convolutional Gated Recurrent Network. A detailed view of our implementation can be found in Appendix.

3.4.2. Application of line RANSAC

We propose a post-processing step after the initial optical flow tracking, utilizing the knowledge that a subset of image points lie on the same line, and therefore also the corresponding image points must lie on a line. Using RANSAC and the line equation, we replace the re-identified points with the intersections of the line equations. During geometric sampling, this step is always used, as it is the basis of the intersection correspondences. Optionally, this step can be used when sampling from the regular grid (see Fig. 5(c)).

3.5. Homography estimation

In this final step, we build a combined set of correspondences $\mathcal{X}_{i+1}$, given by the union of both the keypoint set and the auxiliary set.

$$\mathcal{X}_{i+1} = \mathcal{K}_{i+1} \cup \mathcal{P}_{i+1}$$

Using RANSAC, we estimate the homography H_{i+1}.

Table 2 shows the error incurred when using a homography to model the SoccerNet GSR ground truth data, quantified by Location Similarity and median distance in world coordinates. Ground truth image points are projected using a homography derived by the pitch markings. As we believe the observed errors to be negligible, we do not model non-linear distortions explicitly.

4. Experimental setup

4.1. Dataset

Our experiments use the SoccerNet Game State Reconstruction (SoccerNet-GSR) dataset (Somers et al., 2024), a resource for tracking athletes on a 2D pitch representation. It provides 200 clips of 30-second broadcast sequences captured by a single, moving camera, so only a portion of the pitch is visible at any given time. The dataset is split into training, validation, test, and challenge sets.

Extending the SoccerNet-Tracking dataset, SoccerNet-GSR adds annotations crucial for player localization, including over 9.37 million field marking annotations and 2.36 million athlete pitch positions. The dataset assumes a standard pitch size of 105×68 meters. Ground truth athlete positions are derived by projecting the bottom-centre point of their bounding box onto the pitch plane using the provided camera calibration.

(a) Patch 1 (b) Patch 2 (c) Flow Vectors

Fig. 7. Example of the Patch dataset. A data point consisting of two patches (Fig. 7(a), 7(b)) from neighbouring frames taken from the same image coordinates. The flow field (Fig. 7(c)) is calculated using the procedure described in Section 4.1. Best viewed on screen.

Some of our experiments required the curation of specific subsets of the GSR dataset. These datasets are outlined below.

HSB dataset. To evaluate the propagation method separately from anchor frame selection, we devise the Hard Sequences Benchmark (HSB) dataset. Ten sequences from test set videos of the GSR dataset were manually chosen based on poor LocSim scores using NBJW (see Table 3). The sequences involve camera zoom, high motion or tilt over regions with few detectable keypoints.

Segmentation dataset. To train the U-Net segmentation model used for the surrogate metrics (see Section 3.5), we use all images from the SoccerNet GSR dataset and derive binary masks based on the ground truth homography H_i, acquired using RANSAC over all annotated correspondences. A slight adjustment was made to the data split in the segmentation dataset, moving videos 41–49 from validation to train split, as large shadows on the pitch were over-represented in the validation set. Therefore, the dataset comprised 49,836 training images, 35,334 validation and 35,509 test images.

Optical flow dataset generation. To generate a dataset suitable for learning optical flow, we extract square image regions of size 80×80 pixels from player-free areas, ensuring that motion within these regions arises solely from camera movement. This enables the annotation of dense pixel correspondences using the ground-truth homographies associated with each frame. Each data sample consists of two corresponding image regions from consecutive frames, cropped at identical coordinates. Given two consecutive frames f_i and f_{i+1} with homographies H_i and H_{i+1}, pixel-wise correspondences are obtained by transforming image coordinates according to $\mathbf{x}'_{i+1} = \mathrm{H}_i \mathrm{H}_{i+1}^{-1} \mathbf{x}'_i$. The resulting displacement vectors between corresponding coordinates define the optical flow field. For each frame pair, six such regions are sampled using the random point sampling strategy described in Section 3.3.1. In total, this process yields 254,628 samples for the training split and 212,718 for the test split of SoccerNet. The resulting flow annotations constitute surrogate ground truth, as minor homography inaccuracies and subtle photometric variations can induce small deviations from the true optical flow.

4.2. Metrics

To evaluate player-localization performance, which depends directly on the estimated homographies, we adopt the Localization Similarity (LocSim) metric introduced in the GS-HOTA framework (Somers et al., 2024). LocSim quantifies spatial proximity between predicted and ground-truth player positions in world coordinates and is defined as

$$\mathrm{LocSim}(\mathbf{x}_i^P, \mathbf{x}_i^G) = \exp\left(\ln(0.05)\, \frac{\|\mathbf{x}_i^P - \mathbf{x}_i^G\|_2^2}{\tau^2}\right), \qquad (8)$$

Table 3
LocSim scores for selected sequences in HSB dataset using NBJW.

Video ID	117	123	124	128	136	138	150	187	191	193
Sequence	350–500	100–200	250–320	90–0	60–280	130–230	150–280	300–400	450–500	550–650
LocSim using NBJW	0.2784	0.3793	0.6801	0.1578	0.4573	0.4246	0.4462	0.5946	0.1302	0.7025

where $\mathbf{x}_i^P, \mathbf{x}_i^G \in \mathbb{R}^2$ denote predicted and ground-truth world-space locations of player i, and τ is a spatial sensitivity parameter, set to $\tau = 5$ meters unless stated otherwise. This is in line with the original GSR paper (Somers et al., 2024). mAP-LocSim is calculated from $\tau = 1..10m$. Predicted positions $\mathbf{x}_i^P$ are obtained by projecting the lower edge centres of the ground-truth bounding boxes from image space into world space using the estimated homography for each frame.

We involve additional metrics to evaluate our method against other temporally aware methods in Section 2.

To directly assess homography accuracy, we employ the Intersection over Union metric $\mathrm{IoU}_{\mathrm{entire}}$, following Claasen and Villiers (2024). It measures the overlap between the canonical pitch template $\mathcal{T}$ and its projection under the composition of the predicted and ground-truth homographies:

$$\mathrm{IoU}_{\mathrm{entire}} = \frac{\mathcal{T} \cap \mathcal{T}_{\mathrm{proj}}}{\mathcal{T} \cup \mathcal{T}_{\mathrm{proj}}}, \qquad \mathcal{T}_{\mathrm{proj}} = \mathrm{H}_p\, \mathrm{H}_{gt}^{-1}\, \mathcal{T}. \tag{9}$$

A higher $\mathrm{IoU}_{\mathrm{entire}}$ indicates stronger geometric consistency between predicted and ground-truth calibrations. We analogously report $\mathrm{IoU}_{\mathrm{part}}$, which evaluates the overlap after reprojecting the template back into the image domain. For both metrics, we report mean and median.

To further characterize geometric quality, we define projection and reprojection metrics over a regular grid of world points, as used by prior work (Nie et al., 2021; Claasen and Villiers, 2024). Let $\mathcal{G} = \{\mathbf{x}_k \in \mathbb{R}^2\}$ denote a grid sampled every 2.5 m in the x-direction and 2 m in the y-direction of the pitch. These points are projected into the image and distance to the ground truth is measured. The projection can be either a homography, like in our case, or a rich camera parameter model. For a method-specific projection function $\pi_p : \mathbb{R}^2 \to \mathbb{R}^2$, we obtain projected image points $\mathbf{x}'_k = \pi_p(\mathbf{x}_k)$. Normalized Projection Error (NPE) is computed relative to the projection obtained with the ground-truth homography π_{gt}:

$$\mathrm{NPE} = \frac{\|\pi_p(\mathbf{x}_k) - \pi_{gt}(\mathbf{x}_k)\|_2}{h}, \tag{10}$$

where h is the image height. Normalization ensures comparability across resolutions. We report mean and median.

For reprojection, ground-truth projections $\pi_{gt}(\mathbf{x}_k)$ are mapped back to world coordinates using the inverse projection of each method, $\hat{\mathbf{x}}_k = \pi_p^{-1}(\pi_{gt}(\mathbf{x}_k))$, yielding a Reprojection Error (RE) in meters:

$$\mathrm{RE} = \|\hat{\mathbf{x}}_k - \mathbf{x}_k\|_2. \tag{11}$$

Again, we report mean and median values of the RE.

4.3. Implementation details

As the main hardware environment, we use a workstation with an Intel i7-8700K and an NVIDIA RTX 4090 GPU. Our code is built in Python 3.9 with PyTorch 1.13 and CUDA 12. The pipeline is based on the SoccerNet GSR Challenge Repository, in itself a fork of Tracklab (Joos et al., 2024).

5. Experiments

We design our experiments to evaluate the core components of our pipeline, validate our specific contributions, and benchmark against State-of-the-Art methods. First, Section 5.1 uses the smaller HSB dataset to quantify propagation performance, analysing the influence of point sampling, the contributions of both point sets, and optical flow estimation. Next, we evaluate the complete pipeline with automatic anchor frame initialization on the test split of the SoccerNet GSR dataset, ablating anchor selection and auxiliary point sampling in Sections 5.2 and 5.3, respectively. To demonstrate the improvements of AuxFlow qualitatively, we compare it to NBJW frame-by-frame in Section 5.4. Runtime analysis including data for each component is given in Section 5.5. Finally, AuxFlow is benchmarked against comparable temporally aware methods in Section 5.6.

Table 4
Comparison of methods on the HSB dataset using H_{gt} for initialization. Sampling and re-identification methods are varied. The mean and median values of the LocSim metric are reported.

Method	Sampling strategy	Re-identification	LocSim mean (%) ↑	LocSim median (%) ↑
NBJW	only $\mathcal{K}_{i+1}$ used	–	43.15	35.93
Ours	Regular Grid	RAFT	78.71	91.73
Ours	Regular Grid	no flow	23.42	0.50
Ours	Random	Lucas–Kanade	83.39	95.23
Ours	Geometric	Lucas–Kanade	39.68	17.68
Ours	Regular Grid	Lucas–Kanade	**83.67**	**96.15**
Ours	Regular Grid + Line RANSAC	Lucas–Kanade	82.84	95.71
Ours	Regular Grid, only $\mathcal{P}_{i+1}$, not $\mathcal{K}_{i+1}$ used	Lucas–Kanade	78.30	90.86
Ours	Regular Grid, Bad Initialization	Lucas–Kanade	54.74	68.86

5.1. Propagation performance

5.1.1. Conducted experiment

In this section, we evaluate the propagation performance, isolated from any initialization method, on the HSB dataset, outlined in Section 4.1. This serves to validate the auxiliary point sampling methods described in Section 3.3, assess the effectiveness of the homography propagation approach, and further compare RAFT and Lucas–Kanade re-identification methods. We also benchmark both against a naive baseline that assumes no visual flow. Furthermore, we evaluate the impact of excluding either set of points, using only $\mathcal{K}_{i+1}$ (the NBJW baseline), or using only $\mathcal{P}_{i+1}$. Ground truth homographies are used for initialization, except in the 'Bad Initialization' result, where we explicitly evaluate the effect of a bad initial homography estimate on the propagation performance. This homograhy is generated by adding random noise of up to 10 percent magnitude to the estimate. For each experiment, we report both the mean and median values of the LocSim metric in Table 4. The key findings in these experiments are summarized below.

5.1.2. Auxiliary point sampling methods

To compare sampling methods, we use Lucas–Kanade optical flow re-identification. Regular Grid sampling performed best, achieving a mean LocSim of 83.67% and a median of 96.15%. Adding line-based RANSAC to this sampling strategy led to a slight performance drop (mean 82.84%, median 95.71%). Random sampling achieved similar performance (83.39% mean), though it was slightly worse in the median. All three approaches substantially outperformed the NBJW baseline, nearly doubling the mean LocSim and tripling the median. Fig. 8(b) shows Regular Grid performance ($\mathcal{P}_{i+1}$+$\mathcal{K}_{i+1}$). Performance

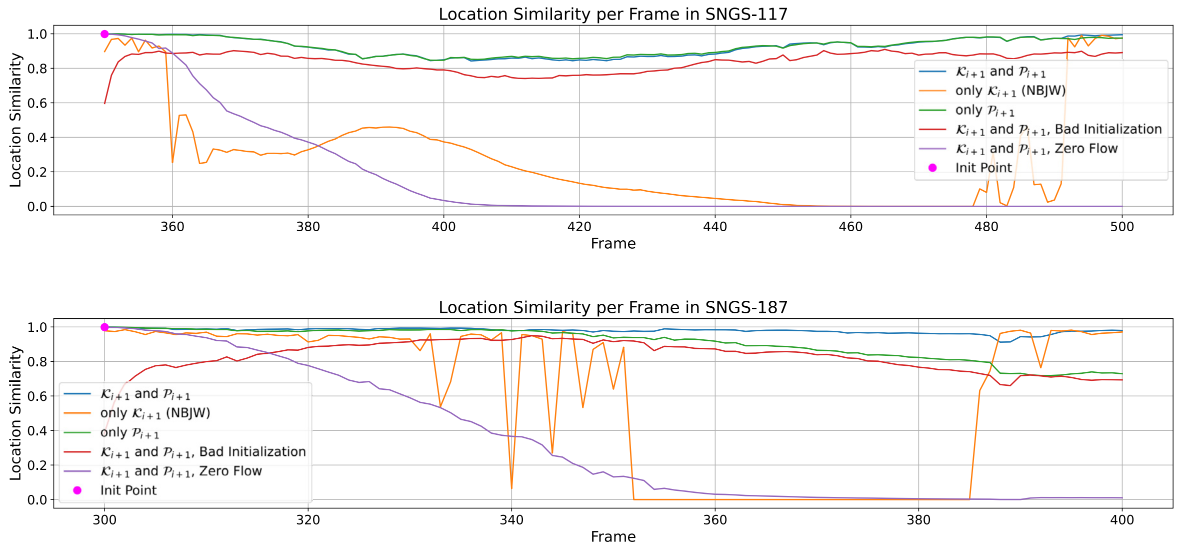


Fig. 8. Visualization of performance on sample sequences in the HSB Dataset. LocSim is displayed per frame. Best viewed on screen.

is clearly superior to NBJW (only $\mathcal{K}_{i+1}$). Interestingly, geometric sampling, despite being more conceptually elaborate, performed poorly, with a mean LocSim of only 39.68% and a median of 17.68%, falling below even the NBJW baseline.

5.1.3. Re-identification methods

Using Regular Grid sampling, we compare Lucas–Kanade and RAFT. An evaluation of the flow estimation itself is shown in Appendix. RAFT achieves higher accuracy in flow estimation than Lucas–Kanade. However, Lucas–Kanade outperformed RAFT in the propagation context across both mean and median LocSim, achieving 83.67% versus 78.71%. This is likely due to Lucas–Kanade retaining far more points when utilizing forwards–backwards tracking. Seemingly, homography estimation benefits more from a higher number of correspondences than fewer highly-accurate ones. Both methods significantly outperformed the naive approach that assumes zero flow, which yielded a mean LocSim of just 23.42% and a median of 0.50%.

5.1.4. Utilization of $\mathcal{K}_{i+1}$

The propagation step we formulated before utilizes both the keypoints $\mathcal{K}_{i+1}$ estimated by NBJW and the auxiliary points $\mathcal{P}_{i+1}$. In this experiment, we deliberately exclude the keypoint set, which led to a substantial performance drop: mean LocSim decreased to 78.30% and the median fell to 90.86%, underlining their importance in the pipeline to mitigate drift. The effects can be seen in Fig. 8(b). While in SNGS-117 excluding $\mathcal{K}_{i+1}$ did not lead to performance degradation, in SNGS-187 the homography clearly drifts.

5.1.5. Robustness against bad initialization

To assess the reliance on good initial frames, and the behaviour of the propagation mechanism when poor initial estimates are used, we used bad initial homography estimates and measured performance. In both mean (54.74% vs. 43.15%) and median (68.86% vs. 35.93%) LocSim, this approach still outperformed the baseline. Furthermore, observing Fig. 8(b) shows clearly that the propagation can recover from bad initializations, and give good quality estimates in later frames.

5.1.6. Discussion

The experimental results collectively validate the proposed homography propagation framework, highlighting a critical trade-off between correspondence quantity and individual feature precision. A central finding is the superior performance of Regular Grid sampling combined with Lucas–Kanade re-identification. While RAFT demonstrates higher raw flow accuracy, the homography estimation process benefits significantly more from the high density and retention rate of points provided by Lucas–Kanade than from the sparser, albeit more accurate, flow vectors of RAFT. This suggests that for planar tracking, the statistical robustness gained from a large number of uniformly distributed correspondences outweighs the benefits of high-precision deep learning-based flow estimation.

Furthermore, the poor performance of geometric sampling indicates that complex, feature-dependent heuristics may introduce instability or filter out useful contextual data in this domain. A simple, uniform spatial coverage – as provided by the Regular Grid – proves most effective in constraining the homography, likely by minimizing the risk of degenerate configurations in texture-poor regions.

Finally, the ablation studies regarding $\mathcal{K}_{i+1}$ and initialization quality underscore the system's hybrid resilience. The performance degradation observed when excluding $\mathcal{K}_{i+1}$ confirms that while auxiliary points $\mathcal{P}_{i+1}$ are essential for smooth short-term propagation, the external keypoints act as a necessary anchor to mitigate long-term drift. The system's ability to recover from bad initializations further corroborates this self-correcting behaviour, demonstrating that the fusion of dense auxiliary tracking and sparse keypoint detection creates a robust pipeline capable of maintaining high localization performance even under suboptimal starting conditions.

5.2. Ablation of anchor frame selection

5.2.1. Experiment description

We examine all three proposed anchor frame selection approaches detailed in Section 3.2, using the SoccerNet GSR datasets test split. Regular Grid Sampling and Lucas–Kanade optical flow are used. Anchor frames are those where the respective metric is locally maximal across frames. Moreover, we use the heuristic in combination with either one of the surrogate metrics. Now, an anchor frame is a local maxima of the metric across frames that also fulfils the heuristic. Finally, we will select anchor frames naively by simply using every 150th frame, therefore gauging the importance of anchor frame selection. As an upper bound, we will perform selection using the true $\mathrm{IoU}_{\mathrm{entire}}$ values of H_i, giving us an understanding of how much performance is lost due to our selection strategies.

Table 5
Ablation of our pipeline using the LocSim metric. Anchor frame selection and sampling methods are varied.

Optical flow	Anchor frame selection	Sampling	LocSim (%) ↑
LK	naive	Regular Grid	82.483
LK	Heuristic	Regular Grid	88.463
LK	Image U-Net	Regular Grid	90.486
LK	Pitch U-Net	Regular Grid	90.074
LK	Heuristic + Image U-Net	Regular Grid	90.647
LK	Heuristic + Pitch U-Net	Regular Grid	90.075
LK	Ground Truth (IoU_{entire})	Regular Grid	91.879
LK	Heuristic + Image U-Net	Geometric	86.149
LK	Heuristic + Image U-Net	Random	89.900
LK	Heuristic + Image U-Net	Regular Grid	90.647
LK	Heuristic + Image U-Net	Regular Grid + Line RANSAC	89.617

5.2.2. Results

The upper section of Table 5 shows the result of this experiment. Naively initializing the propagation every 150 frames leads to performance degradation (82.483%) relative to the NBJW baseline (83.481%). By contrast, all proposed initialization strategies yield improvements. The keypoint heuristic achieves 88.463%, and surrogate metrics yield further gains: 90.486% for image-based U-Net IoU and 90.074% for the pitch-based counterpart. Combining the keypoint heuristic with the image-based surrogate produces the best performance (90.647%). As an upper-bound, initialization based on the true IoU_{entire} of NBJW predictions results in 91.879%.

5.2.3. Discussion

We were able to devise multiple selection methods, all of which enabled the pipeline to improve upon NBJW. Both the heuristic approach and the surrogate metrics performed well. In fact, taking into account the upper bound based on anchor frames chosen with IoU_{entire} scores, all approaches were within the last 20% of theoretically possible performance uplift over NBJW. Concluding, a combined selection based on heuristic and image-space surrogate metric performed best.

5.3. Ablation of point sampling

5.3.1. Experiment description

Using the best performing initialization strategy from the previous section (Heuristic + Image-based U-Net surrogate), we further examine the performance differences of the sampling methods. Otherwise, this experiment follows the same principles as the previous ablation.

5.3.2. Results

Results are summarized in Table 5. Regular Grid once again performs best (90.647%). Grid + Line RANSAC (89.617%) and Random Points (89.900%) are slightly worse, but still significantly outperform NBJW. The Geometric sampling approach yields modest improvement (86.149%) but remains well below the best-performing methods.

5.3.3. Discussion

While random and regular grid sampling strategies outperformed NBJW in both the HSB and pipeline ablation experiment, geometric sampling only slightly outperformed NBJW in the pipeline experiment, while failing completely on the HSB dataset. The key idea behind this sampling strategy was to use intersections of defined lines as correspondences, rather than simple points, and re-identify these lines through many tracked samples along each line using RANSAC. Results suggest that while sampling in such a manner is sufficient for temporal smoothing, propagation across regions of poor homography accuracy does not yield desired performance. It seems, defining intersections of lines as correspondences does not yield high enough accuracy to outweigh the downside of fewer correspondences when using RANSAC. Post-processing the Regular Grid Sampling using line RANSAC degraded homography estimation slightly, although not by a large margin. Concluding, propagation works best when sampling auxiliary points distributed regularly on the pitch.

Table 6
Runtime evaluation of the AuxFlow pipeline. For non-GPU tasks, batchsize shows the number of threads.

Component	Batchsize	Runtime (s)	FPS (1/s)
NBJW keypoints	8	54	13.9
NBJW H estimation	1	27	27.8
U-Net initialization	1	61	12.3
Propagation	1	150	5
Overall	–	292	2.6

5.4. AuxFlow qualitative results

Based on the evaluation of previous experiments, we have determined that the optimal configuration for overall performance involves regular grid sampling for auxiliary points, the Lucas–Kanade method for re-identifying those points, and a combination of heuristic and image space IoU surrogate for anchor frame selection. From now on, we will denote this combination AuxFlow. Now, we review AuxFlow closely, and find both subtle improvements in parts of the video where NBJW was also accurate, and considerable gains when NBJW begins to fail, see Fig. 9. Furthermore, we can clearly see AuxFlow's weaknesses and strengths in the shown sequence. In frames 80–130 and 350–500, we see that AuxFlow maintains high localization performance where NBJW fails. These two sequences, as well as many more such sequences in the dataset, are characterized by light to medium camera motion or apparent motion due to zoom, and non-extreme motion blur. When encountered with extreme, abrupt camera movement or motion blur, like in frames 560–610, AuxFlow cannot gain significantly on NBJW. Furthermore, in sequences of very high NBJW performance, such as frames 130–350, AuxFlow is still able to considerably outperform it due to the temporal smoothing aspect implicitly formulated in our algorithm.

5.5. Evaluation of runtime

To quantify computational cost, we evaluate the runtime of AuxFlow in Table 6. Each component's runtime is individually listed, as well as the overall result. AuxFlow currently runs at 2.6 FPS, largely due to poor optimization, the use of the development framework Tracklab (Joos et al., 2024), and use of Python instead of more performant languages. Specifically, the propagation module acts as the primary bottleneck, consuming over half of the total processing time. Transitioning this component to a compiled language like C++, or optimizing its batch processing, would significantly reduce latency. Despite the current overhead, the system remains viable for offline analysis where high-fidelity tracking takes precedence over real-time constraints.

5.6. Comparison to other temporally aware methods

For further inspection, we evaluate AuxFlow, as presented in the previous chapter, against other temporally aware methods from the literature. The test split of the SoccerNet GSR dataset is used.

5.6.1. Competing methods

Claasen and Villiers (2024) use a two stage Kalman filter on both the estimated keypoints and the estimated homography to great effect on the TSWC homography estimation dataset (Chu et al., 2022). NBJW however was shown to outperform their method, most likely due to the less performant keypoint model used in BHITK. The original implementation is not accessible. To assess the applicability of temporal filters,

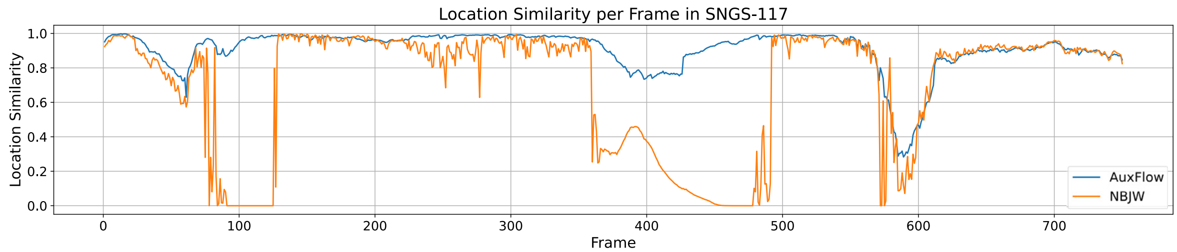


Fig. 9. Our AuxFlow method (blue) against NBJW method (orange) over all 750 frames of video SNGS-117. Both subtle improvements in regions of good NBJW performance and significant gains in regions of very bad NBJW performance can be observed. Best viewed on screen.

Table 7
Performance evaluation against State-of-the-Art competing methods. All metrics are gathered per frame, and median and mean are then calculated. An upwards arrow behind the metrics indicate that higher values correspond to better performance. Likewise, a downwards arrow indicates lower values to be better.

Method	IoU_{entire} ↑		IoU_{part} ↑		NPE ↓		RE (m) ↓		LocSim ↑		
	mean	median	mean	median	mean	median	mean	median	$\tau = 5$ m	$\tau = 2$ m	mAP
NBJW	0.7980	0.8647	0.9288	0.9686	0.0199	0.0143	0.9243	0.5457	0.8851	0.6995	0.9482
NBJW + BHITK	0.7913	0.8594	0.9290	0.9669	0.0202	0.0150	0.9575	0.5669	0.8852	0.6950	0.9494
BroadTrack	**0.8397**	**0.8908**	0.9224	0.9520	0.0320	0.0254	1.3381	0.9302	0.8597	0.5929	0.9431
AuxFlow (Ours)	0.8352	0.8853	**0.9426**	**0.9720**	**0.0186**	**0.0134**	**0.8410**	**0.5289**	**0.9290**	**0.7548**	**0.9751**

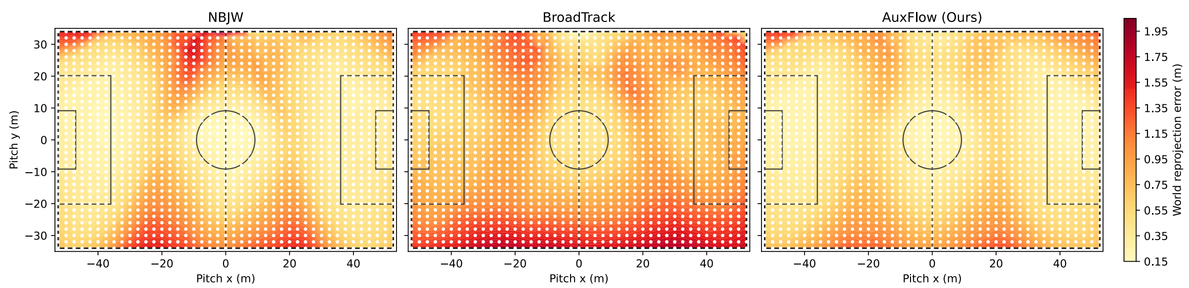


Fig. 10. Median REs obtained from regularly sampling pitch coordinates (as indicated by regular bright spots in all three images) and computing the distance to their reprojected counterparts. Reprojection was achieved by projecting them to image coordinates using ground truth homographies and reprojecting these back to the pitch using the projection function as estimated by NBJW (left), BroadTrack (middle), and AuxFlow (right). Best viewed on screen.

we implemented the filters to use NBJW for keypoint estimation. This will be denoted as NBJW + BHITK. We checked our implementation on the TSWC dataset, where it slightly improved performance over NBJW, consistent with the results claimed by the authors.

Furthermore, we compare against BroadTrack (Magera et al., 2025), a recently published method reporting SOTA performance on the SoccerNet GSR dataset. It computes camera parameters for every frame by minimizing a loss function that integrates reprojection errors from field markings, optical flow correspondences, and the physical constraints of a tripod model. The methods' implementation is that of the authors, freely available on GitHub.[2]

5.6.2. Results

The quantitative results, presented in Table 7, demonstrate that our method achieves superior performance across the majority of evaluation metrics.

In terms of homography accuracy, BroadTrack achieved the highest performance, with a mean IoU_{entire} of 0.8397. However, our method (AuxFlow) remained highly competitive in this category (mean: 0.8352) while significantly outperforming all other methods in the part-based IoU metric. AuxFlow achieved the highest IoU_{part} with a mean of 0.9426 and a median of 0.9720.

Our method demonstrated robust geometric accuracy, recording the lowest errors among all evaluated frameworks. AuxFlow achieved the best Normalized Projection Error (NPE) (mean: 0.0186) and Reprojection Error (RE) (mean: 0.8410 m). Notably, while BroadTrack performed well in IoU metrics, it exhibited a significantly higher mean RE (1.3381 m), suggesting that while its mapping of the pitch outline is accurate, its underlying geometric calibration may be less precise than that of AuxFlow.

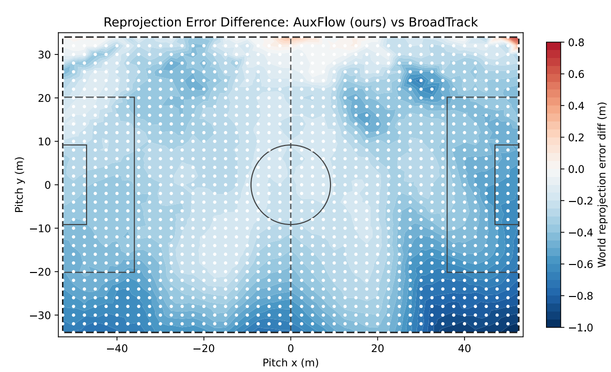


Fig. 11. Difference in RE per field grid point between AuxFlow and Broadtrack. Blue regions show AuxFlow outperforming BroadTrack. Best viewed on screen.

AuxFlow consistently outperformed competing methods in localization tasks across all thresholds. It achieved the highest LocSim scores

[2] https://github.com/evs-broadcast/BroadTrack

for $\tau = 5m$ (0.9290) and $\tau = 2m$ (0.7548), as well as the highest mean Average Precision (mAP) of 0.9751. This indicates that our method provides the most reliable player localization capabilities.

The addition of BHITK to the NBJW keypoint model resulted in negligible differences. The performance metrics for NBJW + BHITK (e.g., mean $\text{IoU}_{\text{entire}}$ of 0.7913) were similar to the standard NBJW (mean $\text{IoU}_{\text{entire}}$ of 0.7980), indicating that the addition did not meaningfully improve or degrade performance in this context. We believe due to the longer sequences and more heterogeneous data in the GSR dataset, the Kalman filters were not able to show the same uplift in accuracy as reported by their authors.

5.6.3. Error distribution

To investigate the spatial distribution of errors, Fig. 10 presents the median REs across the pitch for NBJW, BroadTrack, and AuxFlow. Although NBJW performs well within the penalty box and centre circle, performance degrades significantly in intermediate zones and along the perimeter. BroadTrack exhibits elevated error levels across the entire field. Conversely, AuxFlow maintains consistently lower errors without the localized spikes observed in NBJW. Additionally, Fig. 11 highlights the performance differential between BroadTrack and AuxFlow, with blue regions denoting areas where our method demonstrates superior accuracy.

This analysis underscores NBJW's dependency on keypoint availability, as errors escalate in areas lacking distinct pitch markings. Furthermore, BroadTrack's underperformance suggests a limitation in its optimization strategy, which focuses on minimizing projection error relative to field markings while neglecting the holistic representation of the pitch surface.

6. Discussion and conclusion

In this study, we present AuxFlow, a novel approach to video-wise field registration and subsequent player localization for soccer broadcast footage. By extending a State-of-the-Art frame-based keypoint prediction model through sampling, projection, and re-identification of auxiliary pitch points, we enable homography re-estimation that leverages temporal and visual consistency across frames. This results in improved stability and accuracy over purely frame-wise methods, as shown in our Experiments in Section 5.

We examined the core of AuxFlow, the propagation step, where the auxiliary point set is sampled, projected and re-identified to facilitate homography estimation based on both field keypoints and auxiliary points. Section 5.1 shows this works to great effect using the HSB dataset. Further, both sets of points were shown to be key to achieving the best performance, validating our approach. Flow estimation for re-identification was also shown to be integral to mitigate drift. Lastly, propagation shows robustness towards bad initial homography estimates, a very encouraging result.

Evaluating the entire pipeline on the SoccerNet GSR dataset, we ablated anchor frame selection (cmp. Section 5.2) and auxiliary point sampling strategies (cmp. Section 5.3). While all devised selection methods proved viable, the combination of keypoint heuristic as a pre-selection of potential anchor frames and the image-space IoU surrogate metric for final selection resulted in best performance. In fact, it achieves 80% of the theoretically possible performance uplift over NBJW using our propagation pipeline, gauged by selection via ground truth knowledge. Sampling auxiliary points with a regular field grid gave the best results, with random and geometric sampling performing slightly worse. Line RANSAC refinement of the re-identified points slightly degraded performance.

These ablation studies enabled us finding the best performing methods for each module of our AuxFlow pipeline. In Section 5.4, we compared it frame-to-frame to the NBJW keypoint method. In zoomed views, or when slow to moderate camera movement is present, AuxFlow shows huge gains over NBJW. In high motion or motion blurred scenes, performance drops to negligible uplift over NBJW. This, however, still results in a high overall improvement.

To ascertain real time viability, we examined the computational cost of AuxFlow. It shows large room for improvement, as the current implementation only achieves 2.6 FPS. Further analyses and optimizations may lead to a more performant system.

As detailed in Section 5.6, a comparative analysis with BroadTrack and BHITK highlights the clear superiority of our method. Although BroadTrack attains a slightly higher $\text{IoU}_{\text{entire}}$, AuxFlow outperforms competing methods in every other metric, underscoring its precise geometric representation (NPE and RE) and player localization (LocSim). Ultimately, our experiments show that inducing temporal context, both through auxiliary point sampling and propagation, and through anchor frame selection, significantly enhances SOTA keypoint model performance. Consequently, AusFlow proves superior to existing State-of-the-Art methods. We provide full access to our code to facilitate reproducibility and to encourage future work.

Despite these strengths, the approach carries several limitations. The initialization relies on heuristic and surrogate metrics which, although highly effective in our experiments, may depend on dataset-specific properties. The homography model assumes a pinhole projection without non-linear distortion parameters. While our analyses suggest that this simplification does not hinder performance on the evaluated data, it may become limiting when applied to footage with stronger lens effects or non-standard camera setups. In these cases, it should prove straightforward to compensate these effects. Furthermore, the annotations of the SoccerNet GSR dataset assume homogeneous pitch dimensions across stadiums, which may lead to projection inaccuracies due to labels inconsistent with geometric realities.

These considerations point towards several promising directions for future work. The initialization heuristics could be incorporated into a dynamic control mechanism, allowing the system to self-correct during propagation. Explicit modelling of non-linear distortion should be explored to improve robustness across diverse datasets. Recent end-to-end transformer-based architectures, such as VGGT (Wang et al., 2025), offer an opportunity to embed the inductive biases used here (temporal consistency, geometric interpretability, and projection structure) into a unified trainable pipeline. Such a model could generalize more broadly and require minimal manual tuning when transferred to new domains. Finally, relaxing the assumption of fixed pitch dimensions by leveraging geometric invariants, combined with partial field layout priors, could enable automatic estimation of field scale and geometry at inference time. This capability would support fully calibration-free deployment and extend applicability beyond soccer. Although outside the present scope, we expect the proposed methodology to transfer well to other team sports with sufficiently structured playing surfaces, a hypothesis that warrants dedicated evaluation and the development of new datasets.

CRediT authorship contribution statement

Julian Ziegler: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniel Matthes:** Visualization, Validation, Formal analysis. **Patrick Frenzel:** Validation, Methodology, Investigation, Data curation. **Mirco Fuchs:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 8
Performance comparison of point re-identification models. For further information, see Appendix. An upwards arrow behind the metrics indicate that higher values correspond to better performance. Likewise, a downwards arrow indicates lower values to be better.

Re-identification method	Avg pixel error (px) ↓	PCK@3px ↑	PCK@5px ↑	PCK@8px ↑	Points detected (%) ↑
Predict Mean	8.00	0.34	0.48	0.63	–
Lucas–Kanade	6.38	0.51	0.65	0.77	–
Lucas–Kanade w/ f-b-check ($\varphi = 1$ px)	3.89	0.60	0.76	0.87	**80.82**
RAFT	5.23	0.55	0.70	0.82	–
RAFT w/ f-b-check ($\varphi = 1$ px)	**3.05**	**0.73**	**0.84**	**0.91**	21.54
RAFT w/ f-b-check ($\varphi = 2$ px)	3.27	0.72	0.83	0.91	27.44
RAFT w/ f-b-check ($\varphi = 3$ px)	3.41	0.71	0.83	0.90	32.72

Appendix. Optical flow re-identification results

We evaluate both the Lucas–Kanade and our RAFT implementation. These methods serve the purpose of re-identifying the auxiliary points in consecutive frames, enabling us to establish new correspondences.

First, we evaluate the RAFT model on the test split of the patch dataset, alongside the Lucas–Kanade method. Results are given in Table 8. Additionally, a naive baseline that predicts the mean of the dataset is included for performance comparison.

The average pixel error, as well as percentage of correct keypoints (PCK) at 3, 5, and 8 pixels from the ground truth, are reported. Both methods performed significantly better than simply predicting the mean. RAFT achieved an average pixel error of 5.23, while Lucas–Kanade scored 6.38. This validates both approaches in a first instance.

Utilizing the forward–backward check, as described in the previous chapter, further improved the performance of both methods. Lucas–Kanade, with φ set to 1 pixel, achieved a 3.89 average pixel error, 0.60 PCK@3, 0.76 PCK@5, and 0.87 PCK@8. Furthermore, 80.82% of the points were retained, with only about a fifth being rejected due to forward–backward inconsistencies.

RAFT, with forward–backward checking, was evaluated with three different thresholds φ (1, 2, and 3 pixels). All three configurations outperformed standalone RAFT and both Lucas–Kanade variants. As expected, $\varphi = 1$ px yielded the highest accuracy, with an average pixel error of 3.05. Across all thresholds, the accuracy metrics remain comparable: for example, an average pixel error of 3.41 was observed at a 3-pixel threshold. Similarly, all PCK values remained within a narrow range (±2%). However, the proportion of retained points varies significantly with threshold. Using a 1-pixel threshold retained only 21.54% of points, while setting φ to 3 pixel retained 32.72%, an increase of nearly 50%.

In summary, testing re-identification methods independently reveals a clear trade-off: RAFT with forward–backward checking yields the best accuracy, whereas Lucas–Kanade achieves similar performance while retaining more than twice as many points. Without further evaluation, it is unclear which method is more suitable for our specific use case.

Data availability

Data will be made available on request.

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